CVS · DAY 2 · PUMPING MECHANICS

USMLE Step 1 · June Cohort

PREPARED BY ALLAN BAKESIGA · USMLE STEP 1 PREP

bakesiga.github.io/usmle-dashboard

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CO ARITHMETIC

FICK · EF · CO

- $CO = SV \times HR$ $SV = EDV - ESV$
- $EF = SV / EDV$ — normal ≥ 50%; HFrEF < 40%.
- Fick: $CO = VO_2 / (CaO_2 - CvO_2)$ — used when thermodilution unreliable.
- $MAP \approx DBP + \frac{1}{3}(SBP - DBP) \approx CO \times SVR$
- **Pulse pressure** = SBP − DBP — determined by SV and aortic compliance.
- Worked: EDV 125, ESV 75, HR 100 → SV 50, EF 40%, CO 5 L/min.
- Coronary sinus = **lowest SpO_2 in the body** (~30%); heart extracts >70% O_2 at rest → demand met by ↑ flow only.

HEMODYNAMICS

OHM · POISEUILLE · PP

- $\Delta P = Q \times R$ → vasopressors raise SVR; nitrates / hydralazine drop it.
- $R \propto \eta L / r^4$ — halving radius raises resistance **16×**. Arterioles control BP.
- **Wide PP**: aging aorta (ISH), AR, anemia, pregnancy, hyperthyroid, AV fistula.
- **Narrow PP**: cardiogenic shock, severe AS, tamponade.
- Capillary Starling edema: ↑ Pc (HF), ↓ πc (nephrotic / liver / malnutrition), ↑ permeability (sepsis, burns, anaphylaxis), lymphatic block (filariasis).

PRELOAD · AFTERLOAD · CONTRACTILITY · FRANK-STARLING

BEDSIDE MANEUVERS

- **Preload** ≡ EDV (venous return). **Afterload** ≈ aortic DBP. **Contractility** = force at fixed preload.
- **Frank-Starling**: ↑ EDV → ↑ sarcomere length (~2.2 µm optimal) → ↑ cross-bridges + Ca-sensitivity → ↑ SV.
- ↑ Contractility: catecholamines (β_1), digoxin (Na/K ATPase block → ↑ cytosolic Ca via NCX), ↑ extracellular Ca, thyroid hormone.
- ↓ Contractility: β-blockers, non-DHP CCB, acidosis, hypoxia, ischemia, HFrEF.

Maneuver	Preload	Afterload	Louder	Softer
Valsalva strain / standing	↓	↓ then ↑	**HCM, MVP**	AS, MR, AR
Squat / passive leg raise	↑	↑	AS, MR, AR	**HCM** (MVP click later)
Sustained handgrip	—	↑	AR, MR, VSD	**AS, HCM**
Inspiration (Carvallo)	↑ R-side	—	TR, TS, PR, PS	(L-sided)

PRESSURE-VOLUME LOOP · NORMAL LANDMARKS

PV LOOP = DAY 2 CENTERPIECE

- **Point 1** MV closes (S1) → **1→2** isovolumetric contraction → **Point 2** AV opens.
- **2→3** ejection → **Point 3** AV closes (S2) → **3→4** isovolumetric relaxation.
- **Point 4** MV opens (*opening snap if MS*) → **4→1** diastolic filling.
- **Loop width = SV.** Top-left line = **ESPVR** (steeper = ↑ contractility). Bottom curve = **EDPVR** (up-shift = stiff, non-compliant LV).

SIX PV-LOOP PERTURBATIONS

MEMORIZE ALL SIX

Perturbation	EDV	ESV	SV	Peak P	Notes
↑ **Preload** (saline, AV fistula)	↑	—	↑	slight ↑	Wider loop, shifts right.
↑ **Afterload** (phenylephrine, AS)	—	↑	↓	↑↑	Taller, narrower loop; ↑ wall stress.
↑ **Contractility** (dobutamine, exercise)	—	↓	↑	↑	Steeper ESPVR slope.
AV fistula	↑	↓	↑↑	—	↑ preload + ↓ afterload → chronic high-output state.
Nitroglycerin (venodilator)	↓	↓	↓	↓	Narrow loop → ↓ wall stress (antianginal).
Nitroprusside (balanced)	↓	↓	**preserved**	↓	Drug of choice in acute HF + HTN.

WIGGERS + JVP WAVES
A C X V Y

- **a** = atrial systole — *lost in AFib*; **cannon a** = AV dissociation (complete heart block).
- **c** = TV bulge during isovolumetric contraction.
- **x descent** = atrial relaxation + RV pulling TV down. Prominent in tamponade.
- **v** = atrial filling against closed TV. **Giant v in TR**.
- **y descent** = TV opens, RA empties rapidly. *Steep in constrictive pericarditis*; blunted in tamponade.
- Mitral inflow Doppler: **E** = early passive filling; **A** = late atrial filling. S3 sits at end of E (sudden deceleration of rapid filling).

HEART SOUNDS · S2 SPLITS
S1 S2 S3 S4

- **S1** = MV+TV closure. Loud in MS (stiff leaflets snap shut); soft in MR.
- **S2** = AV+PV closure (A2 then P2 normally).
- **S3** = early diastolic, rapid passive filling into dilated / volume-overloaded LV. Normal < 40 yo; pathologic in HF.
- **S4** = late diastolic atrial kick into stiff LV. *Always* pathologic in adults — HCM, HFpEF, chronic HTN, AS.

S2 split	Cause
Wide	RBBB, pulmonic stenosis, severe MR
Fixed	**ASD** (RA/LA tied across the shunt)
Paradoxical (split widens on expiration)	**LBBB**, severe AS, HCM

VALVULAR DISEASE · AUSCULTATION + CATH + PV LOOP
FA 316–318

Lesion	Murmur	Pulse / signs	Cath / PV-loop signature
Aortic stenosis	Crescendo-decrescendo systolic, R 2nd ICS, radiates to carotids	**Pulsus parvus et tardus**; triad angina-syncope-HF (5-3-2 y mean survival)	Peak LV » peak aortic; gradient persists through ejection; concentric LVH
Aortic regurgitation	Early diastolic decrescendo, L 3rd–4th ICS (Erb)	**Water-hammer**, wide PP, low DBP; de Musset, Quincke, Müller, Duroziez, Traube, Hill	e.g. aortic 160/60, LV 160/20; wide loop, low DBP, ↑ EDV from regurg return; eccentric LVH
Mitral stenosis	Loud S1 + **opening snap** + mid-diastolic apical rumble (bell, LLD)	Severity tracks **S2–OS interval** (shorter = tighter); AFib, hemoptysis, systemic emboli	↑ PCWP with **normal LV pressures**; LV loop small (under-filled); LA hypertensive
Mitral regurgitation	Holosystolic at apex → axilla	Acute (papillary rupture 3–5 d post-MI): flash pulmonary edema + shock. Chronic: LA dilation, AFib	**No true isovolumetric contraction** phase; loop opens at point 2 below normal; eccentric LVH chronically

CARDIOMYOPATHY ECHO
FA 314–315

Type	Mass	Cavity	EF	Diastole
Dilated	↑	↑	↓	normal
HCM	↑	↓	preserved	impaired
Restrictive	normal	normal	preserved	impaired
Athlete	↑	normal/↑	preserved	normal

- **Concentric** LVH (sarcomeres in parallel) — pressure overload (HTN, AS).
- **Eccentric** LVH (sarcomeres in series) — volume overload (AR, MR, AV fistula, dilated CMP).
- **HCM**: AD; MYH7 / MYBPC3 mutations; asymmetric septal hypertrophy + dynamic LVOT obstruction; S4; murmur *louder* with Valsalva, *softer* with squat / handgrip (opposite of AS); SCD risk on exertion.

EXERCISE + AGING CVS
ISH · LIPOFUSCIN

- **Aerobic exercise**: ↑ HR, ↑ SV, ↑ CO; muscle vasodilation > splanchnic constriction → **net ↓ SVR**. SBP ↑, DBP ≈, PP ↑.
- **Aerobic / volume training** → eccentric LVH (dilated cavity). **Isometric / pressure** → concentric LVH.
- **Aging aorta**: loss of elastin → ↓ compliance → **isolated systolic HTN** (high SBP, normal DBP, wide PP).
- **Aging myocyte**: hypertrophy without hyperplasia; **lipofuscin** (yellow-brown wear-and-tear pigment from lipid peroxidation).
- Aortic valve calcific degeneration → degenerative AS in 70s/80s; mitral annular calcification → mild MR.

TOP 15 TRAPS

1. EF = SV / EDV (not SV / ESV).
2. HCM murmur softer with handgrip — opposite of AS.
3. Fixed split S2 = ASD.
4. Paradoxical split = LBBB or severe AS (split on expiration).
5. MS: normal LV hemodynamics; only the LA is hypertensive.
6. Cath signature of MS: ↑ PCWP with normal LV pressures.
7. Opening snap sits at on the PV loop.
8. AR signs: water-hammer + de Musset + Duroziez + Quincke + Müller + Traube + Hill.
9. Right-sided murmurs louder with inspiration (Carvallo).
10. Posteromedial papillary rupture 3–5 d post-inferior MI → acute MR.
11. Nitroglycerin = venodilator (↓ preload); nitroprusside = balanced (SV preserved).
12. S3 = early diastolic rapid filling; S4 = late diastolic atrial kick.
13. Eccentric LVH = volume overload; concentric LVH = pressure overload.
14. Aging aorta → isolated systolic HTN; pulse pressure widens.
15. Cannon a waves = AV dissociation (complete heart block, junctional rhythm).